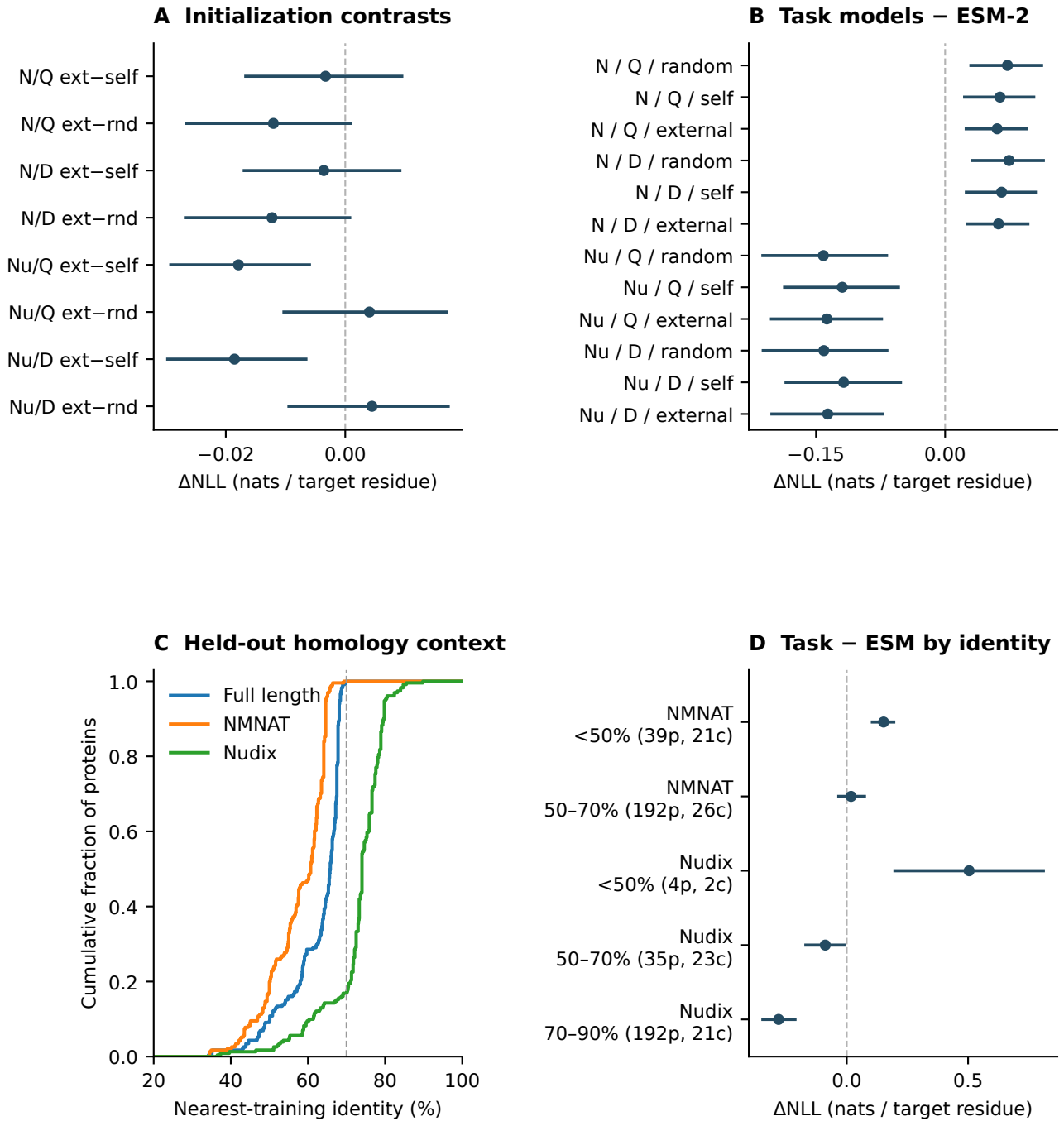
A Initialization contrasts
N/Q ext−self
N/Q ext−rnd
N/D ext−self
N/D ext−rnd
Nu/Q ext−self
Nu/Q ext−rnd
Nu/D ext−self
Nu/D ext−rnd
−0.02
0.00
ΔNLL (nats / target residue)
B Task models − ESM-2
N / Q / random
N / Q / self
N / Q / external
N / D / random
N / D / self
N / D / external
Nu / Q / random
Nu / Q / self
Nu / Q / external
Nu / D / random
Nu / D / self
Nu / D / external
−0.15
0.00
ΔNLL (nats / target residue)
C Held-out homology context
Full length
NMNAT
Nudix
Cumulative fraction of proteins
0.0
0.2
0.4
0.6
0.8
1.0
20
40
60
80
100
Nearest-training identity (%)
D Task − ESM by identity
NMNAT
<50% (39p, 21c)
NMNAT
50–70% (192p, 26c)
Nudix
<50% (4p, 2c)
Nudix
50–70% (35p, 23c)
Nudix
70–90% (192p, 21c)
0.0
0.5
ΔNLL (nats / target residue)